

NEW DELHI INDIRA GANDHI INTL, India

WMO: 421810

Lat: 28.567N

Lon: 77.103E

Elev: 237

StdP: 98.51

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	7.0	-0.5	3.7	21.0	1.2	4.2	22.4	7.4	18.4	6.5	17.9	1.5	240	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.6	43.8	22.2	42.1	22.3	40.8	22.4	29.2	33.2	28.7	32.6	28.1	31.9	4.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.7	30.8	27.9	24.7	30.6	27.1	23.6	30.1	98.0	32.6	94.8	32.8	92.3	31.8	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.4	5.7	3.3	45.9	1.6	1.6	2.1	47.0	1.2	47.9	0.4	48.8	-0.8	49.9		
			2.8	30.1	1.5	1.2	1.7	31.0	0.9	31.7	0.0	32.3	-1.1	33.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	13.8	17.7	23.4	30.3	33.8	33.9	31.5	30.6	29.9	26.9	21.3	15.6
	DBStd	7.36	2.60	2.97	3.48	3.33	2.96	3.10	2.23	1.68	1.82	2.25	2.42	2.68
	HDD10.0	3	2	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	284	143	44	3	0	0	0	0	0	0	0	4	91
	CDD10.0	5749	119	216	414	608	739	716	666	638	596	524	339	174
	CDD18.3	2988	2	26	159	358	481	466	407	380	346	266	93	6
	CDH23.3	42112	24	248	1807	5344	7843	7483	5797	5166	4527	2992	809	72
	CDH26.7	24982	3	54	836	3386	5458	5123	3351	2732	2328	1476	233	3
Wind	WSAvg	2.5	2.2	2.5	2.6	2.8	3.2	3.3	3.0	2.7	2.4	1.7	1.5	1.7
Precipitation	PrecAvg	583	11	24	11	9	27	54	159	161	108	10	4	5
	PrecMax	983	59	110	90	63	151	154	588	453	301	74	40	50
	PrecMin	234	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	184	15	29	20	15	38	45	134	106	86	21	9	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	31.0	38.0	43.1	45.2	45.8	40.9	38.1	38.0	37.0	32.9	27.0
		MCWB	16.5	18.3	19.1	21.1	21.7	24.0	25.6	24.6	24.2	21.0	18.9	15.9
	2%	DB	23.1	28.1	35.2	41.2	44.0	43.8	39.0	36.9	36.8	35.2	30.8	25.1
		MCWB	14.9	17.2	19.1	20.4	21.5	23.4	26.0	25.6	24.5	21.7	18.4	15.4
	5%	DB	21.2	26.2	33.1	40.0	42.4	42.0	37.2	35.2	35.8	34.2	29.1	23.8
		MCWB	13.8	16.5	18.5	20.1	21.7	23.7	26.8	26.4	24.8	21.6	17.7	14.7
	10%	DB	19.8	24.2	31.1	38.1	41.0	40.1	35.9	34.2	34.2	33.0	27.9	22.0
		MCWB	13.1	15.6	18.2	19.6	21.8	24.3	26.8	26.6	25.3	21.4	17.3	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	20.3	22.3	25.2	27.2	29.3	30.4	29.7	29.1	26.3	21.1	18.0
		MCDB	23.2	26.9	31.3	35.5	37.2	36.2	33.5	33.3	33.5	31.7	27.9	23.4
	2%	WB	16.2	18.9	21.0	23.3	26.1	28.4	29.5	28.9	28.0	25.4	20.0	16.7
		MCDB	20.4	25.2	29.6	35.0	36.5	34.4	33.0	32.3	31.4	30.5	26.6	21.9
	5%	WB	15.2	17.8	20.0	21.9	25.1	27.6	28.9	28.3	27.4	24.5	19.3	15.6
		MCDB	19.2	23.7	29.0	34.9	35.9	33.8	32.6	31.6	31.1	29.7	26.2	21.1
	10%	WB	14.2	16.8	19.2	21.0	24.2	26.9	28.3	27.9	26.9	23.6	18.6	14.7
		MCDB	18.2	22.3	28.1	33.8	35.4	33.9	31.9	31.1	30.7	29.4	25.3	20.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.3	13.5	13.8	12.6	10.1	7.1	6.8	8.3	11.8	13.0	12.4
		MCDBR	13.4	14.0	15.6	15.2	14.1	12.2	9.0	8.3	10.1	12.9	13.8	14.3
		MCWBR	5.9	5.4	4.4	4.6	4.5	3.9	3.2	3.3	3.4	4.0	4.9	6.0
	5% WB	MCDBR	10.9	11.9	13.8	12.9	12.0	9.2	7.4	6.8	7.6	9.6	11.8	12.1
		MCWBR	5.6	5.1	4.6	4.6	4.9	3.9	3.5	3.2	3.3	3.4	4.6	5.7
Clear-Sky Solar Irradiance	taub	0.507	0.482	0.499	0.573	0.670	0.741	0.748	0.653	0.579	0.656	0.630	0.569	
	taud	1.822	1.942	1.931	1.789	1.639	1.555	1.567	1.759	1.878	1.625	1.625	1.703	
	Ebn at Noon	700	776	794	749	679	629	623	681	717	614	583	614	
	Edh at Noon	181	173	186	221	258	279	275	225	194	236	220	196	
All-Sky Solar Radiation	RadAvg	3.03	4.29	5.63	6.61	6.76	5.98	4.93	4.85	5.04	4.60	3.57	3.03	
	RadStd	0.37	0.35	0.26	0.21	0.29	0.44	0.43	0.41	0.44	0.22	0.24	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	+1.05	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page